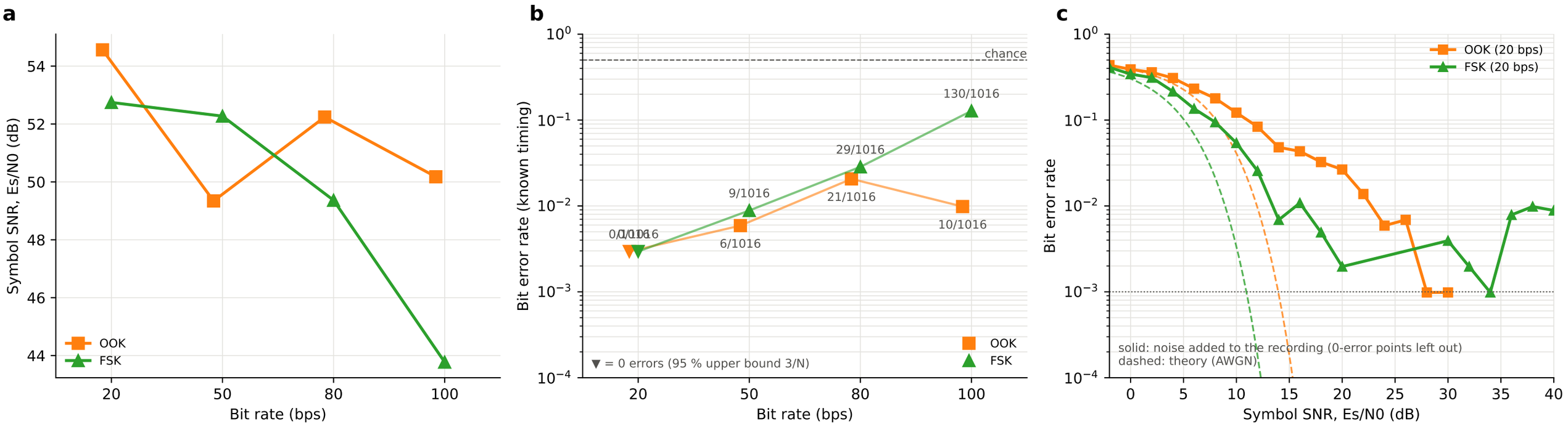
Modulation comparison (1 mA)
a
Symbol SNR, Es/N0 (dB)
54
52
50
48
46
44
20
50
80
100
Bit rate (bps)
OOK
FSK
b
Bit error rate (known timing)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
chance
130/1016
29/1016
9/1016
21/1016
10/1016
6/1016
0/1016
20
50
80
100
Bit rate (bps)
▼ = 0 errors (95 % upper bound 3/N)
OOK
FSK
c
Bit error rate
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
OOK (20 bps)
FSK (20 bps)
solid: noise added to the recording (0-error points left out)
dashed: theory (AWGN)
0
5
10
15
20
25
30
35
40
Symbol SNR, Es/N0 (dB)